

Student Name

Student ID No.:

Section



Computer Engineering Department
Microprocessors Lab - 30 points
Mid Exam

2) Write a program in assembly language to find the total of memory locations that are greater than 10, storing the result into SUM, and push them into the stack.

DB 2,5,8,12,17,1,0,4,13

PCE

INC CX
INC CX
CMP CX,8
JB L

5. MOV AX,27
MOV BL,4
MOV CL,3
DIV BL
XOR AH,FFH

AL= ; AH=

6. Find the content of register AL after execution of the following code:

MOV AL,0FH
MOV CL,2
SHR AL,CL
ADC AL,1

AL=

07110000			
071101: 00	104	1	
071111: 10	01C	1	
071121: 07	007	DEEP	
071131: 02	142	2	
071141: 08	216	2	
071151: 00	176	2	
071161: 25	037	X	
071171: 00	104	1	
071181: 45	067	1	
071191: 23	035	1	
0711A1: 80	137	1	
0711B1: 08	216	1	
0711C1: 0A	138	1	
0711D1: C8	200	1	
0711E1: A0	160	1	
0711F1: 00	000	NULL	
071201: 00	000	NULL	
071211: 01	161	1	
071221: 01	001	0	
071231: 00	000	NULL	
071241: 02	162	6	
071251: 03	003	0	

- A) SI
- B) CS
- C) SS
- D) SP

7. The underlined number represents

8. MOV SI,7
 MOV AL,0
 L2: CMP X[SI],10

AL

87

Multiple Choice Questions: (15) points

1	2	3	4	5	6	7	8	9	10

1. What is the type of addressing modes used in instruction `MOV CL,[BP][SI]`
 A. Indirect B- base relative C- Base Indexed D-Direct Indexed

2. Suppose X defined as below; what is the content of X after execution of the following instructions:

```
X DD 335647FBH,22H
MOV BX,OFFSET X
MOV AL,34H
ADD [BX+1],AL
```

X=

3. Find the content of AL and zero flag (ZF) and carry flag (CF) after execution of the following code:

```
MOV AL,FCH
ADD AL,3
ADC AL,3
```

AL= ; ZF=
CF=

Find the value of AX after execution the following assembler codes

```
MOV CX,1
MOV AX,0
L: ADD AX,CX
INC CX
INC CX
CMP CX,8
JB L
```

AX=

AX,27
AL,4
CL,3

AL= ; AH=

content of register AL after execution of the code:

AL=

```

7  MOV AX, 1020H
   AND AX, 00FFH
   JE YES
   MOV AX, 0
   JMP YES
YES MOV AX, 1
   HLT

```

```

8  MOV BL, 0
   MOV AL, 0
   LD  EBP, 00000010H
   JB  1
   AND AL, 0FFH
   INC BL
   CMP BL, 5
   JB  12
HLT
9  DB 1,00,2,15,00,2,00,07

```

```

10 ORG 100H
    MOV AX, 1020H
    MOV BX, 20H
    PUSH AX
    PUSH BX
    XCHG AX, BL
    ADD AX, BX
    POP AX
    POP BX
    HLT

```

AX=

BX=

10. The result of multiplication instruction MUL BX is stored in register pair:

A: CX:AX

C: CX:AX

Question#2(15 points)

Given memory locations X defined as below

X DB 51,15,76,40,90,45,13,20

- In an assembler codes find the memory locations that are divided by 3;
- Find the sum of them; storing the result into SUM
- Replace them by '0' character
- push them into the stack